**INSFP Médéa**

**Module : Logique Combinatoire**

**Série d'exercices N°1 (Suite)**

**EXERCICE 1 :**

Simplifiez les fonctions suivantes :

$$F_1 = a.(a+b)$$
$$F_2 = (a+b).(\overline{a}+b)$$
$$F_3 = a.b+\overline{c}+c.(\overline{a}+\overline{b})$$
$$F_4 = (x.\overline{y}+z).(x+\overline{y}).z$$
$$F_5 = (x+y).z+\overline{x}.(\overline{y}+z)+\overline{y}$$
$$F_6 = (a+b+c).(\overline{a}+b+c)+a.b+b.c$$
$$F_7 = a+a.b.c+\overline{a}.b.c+\overline{a}.b+a.d+a.\overline{d}$$
$$F_8 = a+\overline{a}.b+\overline{a}.\overline{b}.c+\overline{a}.\overline{b}.\overline{c}.d+\overline{a}.\overline{b}.\overline{c}.\overline{d}.e$$
$$F_9 = (a+b).(a+b.c)+\overline{a}.\overline{b}+\overline{a}.\overline{c}$$

**EXERCICE 2 :**

Simplifiez les fonctions suivantes :

$$F_1 = \overline{a}.b.c+a.c+(a+b).\overline{c}$$
$$F_2 = b.c+a.c+a.b+b$$
$$F_3 = (a.\overline{b}+c).(a+\overline{b}).c$$
$$F_4 = (a.c+b.\overline{c}).(\overline{a}+\overline{c}).b$$
$$F_5 = (\overline{a}.b+a.\overline{b})+(a.b+\overline{a}.\overline{b})$$
$$F_6 = a.b.c+a.b.\overline{c}+\overline{a}.b.\overline{c}+\overline{a}.b.c$$
$$F_7 = a.\overline{b}.\overline{c}+a.b.\overline{c}+a.b.c+a.b.\overline{c}$$
$$F_8 = b.d+c.d+\overline{c}.d+\overline{a}.b.\overline{c}.\overline{d}+a.b.\overline{c}$$
$$F_9 = a.b.c+c.(a.\overline{b}+\overline{a}.b)$$
$$F_{10} = a.b.\overline{c}+b.(a+\overline{c})+\overline{\overline{a}+b+\overline{a}.c}$$

**EXERCICE 3 :**

Complémenter puis simplifiez les fonctions suivantes :

$$T = a.b+b.c+a.c$$
$$F = \overline{c}.\overline{d}+\overline{a}.\overline{b}+c.\overline{d}+a.\overline{b}$$
$$G = \overline{a}.\overline{b}+a.b+a.\overline{b}$$
$$H = \overline{c}.d+\overline{a}.b+c.d+a.b$$

## SOLUTION EXERCICE 1 :

$$F_1 = a.(a+b)$$
$$F_1 = a.a + a.b$$
$$F_1 = a + a.b$$
$$F_1 = a.(1+b)$$
$$F_1 = a.1$$
$$F_1 = a$$

$$F_2 = (a+b).(\bar{a}+b)$$
$$F_2 = a.\bar{a} + a.b + \bar{a}.b + b.b$$
$$F_2 = 0 + a.b + \bar{a}.b + b$$
$$F_2 = b.(1+\bar{a}+a)$$
$$F_2 = b$$

$$F_3 = a.b + \bar{c} + c.(\bar{a}+\bar{b})$$
$$F_3 = a.b + \bar{c} + \bar{a} + \bar{b}$$
$$F_3 = a + \bar{c} + \bar{a} + \bar{b}$$
$$F_3 = 1$$

$$F_4 = (x.\bar{y}+z).(x+\bar{y}).z$$
$$F_4 = (x.\bar{y}+z).(x.z+\bar{y}.z)$$
$$F_4 = x.\bar{y}.x.z + z.x.z + x.\bar{y}.\bar{y}.z + \bar{y}.z.z$$
$$F_4 = x.\bar{y}.z + x.z + x.\bar{y}.z + \bar{y}.z$$
$$F_4 = x.z + \bar{y}.z(1+x)$$
$$F_4 = x.z + \bar{y}.z$$
$$F_4 = (x+\bar{y}).z$$

$$F_5 = (x+y).z + \overline{x}.(\overline{y}+z) + \overline{y}$$
$$F_5 = x.z + y.z + \overline{x}.\overline{y} + \overline{x}.z + \overline{y}$$
$$F_5 = (x+y+\overline{x}).z + \overline{y}.(1+\overline{x})$$
$$F_5 = z + \overline{y}$$

$$F_6 = (a+b+c).(\overline{a}+b+c) + a.b + b.c$$
$$F_6 = a.\overline{a} + a.b + a.c + \overline{a}.b + b.b + b.c + \overline{a}.c + b.c + c.c + a.b + b.c$$
$$F_6 = a.b + a.c + \overline{a}.b + b + b.c + \overline{a}.c + b.c + c + a.b + b.c$$
$$F_6 = (a+\overline{a}+1+c+c+a+c).b + (\overline{a}+1+a).c$$
$$F_6 = b + c$$

$$F_7 = a + a.b.c + \overline{a}.b.c + \overline{a}.b + a.d + a.\overline{d}$$
$$F_7 = a.(1+b.c+d+\overline{d}) + \overline{a}.(b.c+b)$$
$$F_7 = a + \overline{a}.(b.(c+1))$$
$$F_7 = a + \overline{a}.b$$
$$F_7 = a + b$$

$$F_8 = a + \overline{a}.b + \overline{a}.\overline{b}.c + \overline{a}.\overline{b}.\overline{c}.d + \overline{a}.\overline{b}.\overline{c}.\overline{d}.e$$
$$F_8 = a + \overline{a}\left(b + \overline{b}.\left(c + \overline{c}.\left(d + \overline{d}.e\right)\right)\right)$$
$$F_8 = a + b + c + d + e$$

$$F_9 = (a+b).(a+b.c) + \overline{a}.\overline{b} + \overline{a}.\overline{c}$$
$$F_9 = a.a + a.b.c + a.b + b.b.c + \overline{a}.\overline{b} + \overline{a}.\overline{c}$$
$$F_9 = a + a.b.c + a.b + b.c + \overline{a}.\overline{b} + \overline{a}.\overline{c}$$
$$F_9 = a.(1+b.c+b) + b.c + \overline{a}.\overline{b} + \overline{a}.\overline{c}$$
$$F_9 = a + b.c + \overline{a}.\overline{b} + \overline{a}.\overline{c}$$
$$F_9 = a + b.c + \overline{b} + \overline{c}$$
$$F_9 = a + c + \overline{b} + \overline{c}$$
$$F_9 = 1$$

## SOLUTION EXERCICE 2 :

$$F_1 = \bar{a}.b.c + a.c + (a+b).\bar{c}$$
$$F_1 = \bar{a}.b.c + a.(c+\bar{c}) + b.\bar{c}$$
$$F_1 = a + b.(c+\bar{c})$$
$$F_1 = a + b$$

$$F_2 = b.c + a.c + a.b + b$$
$$F_2 = a.c + b(1 + a + c)$$
$$F_2 = a.c + b$$

$$F_3 = (a.\bar{b} + c).(a + \bar{b}).c$$
$$F_3 = a.\bar{b}.c + a.c + .a.\bar{b}.c + \bar{b}.c$$
$$F_3 = a.c(\bar{b} + 1 + \bar{b}) + \bar{b}.c$$
$$F_3 = (a + \bar{b}).c$$

$$F_4 = (a.c + b.\bar{c}).(\bar{a} + \bar{c}).b$$
$$F_4 = \bar{a}.b.\bar{c} + b.\bar{c}$$
$$F_4 = b.\bar{c}$$

$$F_5 = (\bar{a}.b + a.\bar{b}) + (a.b + \bar{a}.\bar{b})$$
$$F_5 = 1$$

$$F_6 = a.b.c + a.b.\bar{c} + \bar{a}.b.\bar{c} + \bar{a}.b.c$$
$$F_6 = a.b.(c+\bar{c}) + \bar{a}.b.(\bar{c}+c)$$
$$F_6 = (a+\bar{a}).b = b$$
$$F_6 = b$$

$$F_7 = a.\bar{b}.\bar{c} + a.b.\bar{c} + a.b.c + a.b.\bar{c}$$
$$F_7 = a.\bar{b}.\bar{c} + a.b.(\bar{c}+c+\bar{c})$$
$$F_7 = a.(\bar{b}.\bar{c} + b) = a.(\bar{c} + b)$$
$$F_7 = a.(\bar{c} + b)$$

$$F_8 = b.d + c.d + \bar{c}.d + \bar{a}.b.\bar{c}.\bar{d} + a.b.\bar{c}$$
$$F_8 = (b + c + \bar{c}).d + (\bar{a}.\bar{d} + a).b.\bar{c}$$
$$F_8 = d + b.\bar{c} + a.b.\bar{c} = d + b.\bar{c}$$
$$F_8 = d + b.\bar{c}$$

$$F_9 = a.b.c + c.(a.\bar{b} + \bar{a}.b)$$
$$F_9 = a.b.c + a.\bar{b}.c + \bar{a}.b.c$$
$$F_9 = a.c.(b + \bar{b}) + \bar{a}.b.c$$
$$F_9 = (a + \bar{a}.b).c = (a + b).c$$
$$F_9 = (a + b).c$$

$$F10 = a.b.\bar{c} + b.(a + \bar{c}) + \overline{\bar{a} + b + \bar{a}.c}$$
$$F10 = a.b.\bar{c} + a.b + b.\bar{c} + a.\bar{b}.\overline{\bar{a}.c}$$
$$F10 = b.(a.\bar{c} + a + \bar{c}) + a.\bar{b}.(a + \bar{c})$$
$$F10 = a.b + b.\bar{c} + a.\bar{b} + a.\bar{b}.\bar{c}$$
$$F10 = a.(b + \bar{b} + \bar{b}.\bar{c}) + b.\bar{c} = a + b.\bar{c}$$
$$F10 = a + b.\bar{c}$$

## SOLUTION EXERCICE 3 :

$$T = a.b + b.c + a.c$$
$$\bar{T} = \overline{a.b + b.c + a.c}$$
$$\bar{T} = (\bar{a} + \bar{b}).(\bar{b} + \bar{c}).(\bar{a} + \bar{c}) \text{ ou } \bar{T} = \bar{a}.\bar{b} + \bar{b}.\bar{c} + \bar{a}.\bar{c}$$

$$F = \bar{c}.\bar{d} + \bar{a}.\bar{b} + c.\bar{d} + a.\bar{b}$$
$$\bar{F} = \overline{\bar{c}.\bar{d} + \bar{a}.\bar{b} + c.\bar{d} + a.\bar{b}}$$
$$\bar{F} = (c + d).(a + b).(\bar{c} + d).(\bar{a} + b)$$
$$\bar{F} = ((a.\bar{a}) + b).((c.\bar{c}) + d)$$
$$\bar{F} = b.d$$

$$G = \bar{a}.\bar{b} + a.b + a.\bar{b}$$
$$\bar{G} = \overline{\bar{a}.\bar{b} + a.b + a.\bar{b}}$$
$$\bar{G} = (a + b).(\bar{a} + \bar{b}).(\bar{a} + b)$$
$$\bar{G} = (a + b).(\bar{a} + (\bar{b}.b))$$
$$\bar{G} = (a + b).\bar{a} = \bar{a}.b$$

$$H = \bar{c}.d + \bar{a}.b + c.d + a.b$$
$$\bar{H} = \overline{\bar{c}.d + \bar{a}.b + c.d + a.b}$$
$$\bar{H} = (c + \bar{d}).(a + \bar{b}).(\bar{c} + \bar{d}).(\bar{a} + \bar{b})$$
$$\bar{H} = ((\bar{a}.a) + \bar{b}).((c.\bar{c}) + \bar{d})$$
$$\bar{H} = \bar{b}.\bar{d}$$